

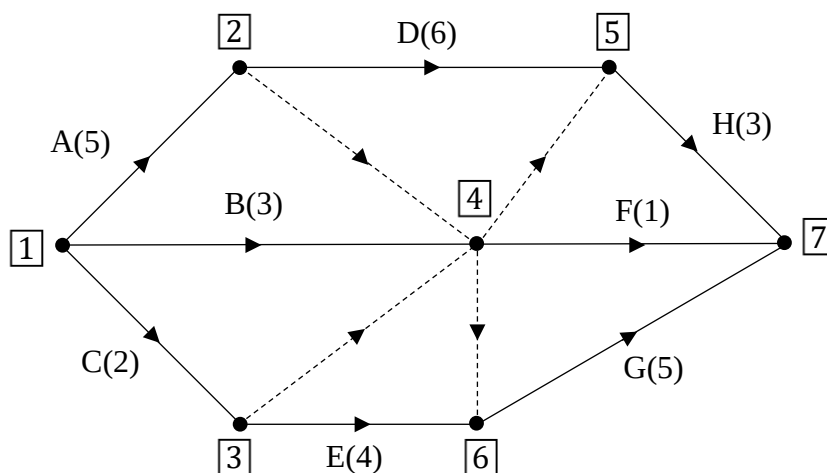
Discrete Mathematics: Activity on arc revision

- 1 The table below shows the activities that need to be completed as part of a task. The immediate predecessors and durations of each activity are shown.

Activity	Immediate predecessors	Duration (hours)
A	-	3
B	-	1
C	-	5
D	A, B	4
E	B, C	6
F	D, E	1
G	B	6

- Represent the above on an activity-on-arc diagram.
- Find the early event time and late event time for each event.
- Give the minimum completion time and list the critical activities.
- Find the float of activity G.

- 2 The activity on-arc diagram below shows the eight activities that need to be completed as part of a project. The duration of each activity is shown in brackets. The events are labelled **1**, **2**, **3**,



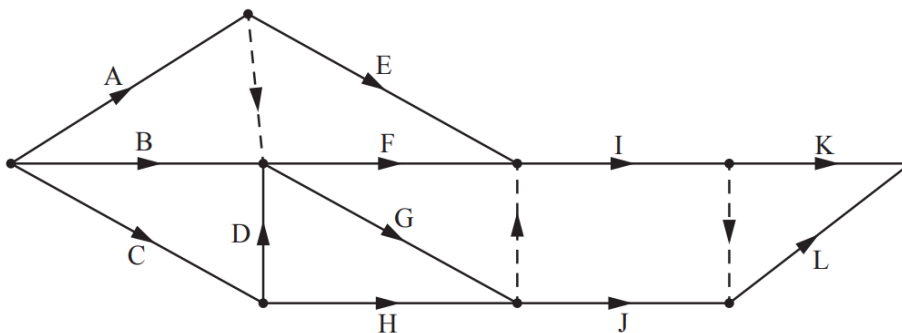
- Produce a table showing the immediate predecessors of each activity.
- Perform a forwards and backwards pass to find the early and late times of each event. Write down the minimum completion time and list the critical activities.

- 3 The table below shows the activities that need to be completed as part of a task. The immediate predecessors and durations of each activity are shown.

Activity	Immediate predecessors	Duration (hours)
A	-	1
B	-	4
C	-	10
D	A	6
E	B	7
F	C	3
G	D, E	5
H	E, F	3
I	G, H	4

- (a) Represent the above on an activity-on-arc diagram.
- (b) Find the early event time and late event time for each event.
- (c) Find the float of each activity, and hence identify the critical activities.

- 4 The network shown below represents a project using activity on arc.



- (a) Explain the reason for the dummy on the left of the diagram.

It is given that activity E is a critical activity. The manager of this project therefore claims that activities A, E, I, and K must form a critical path for the network.

- (b) Is the manager's claim correct? Explain your reasoning.

The durations of the activities are shown below:

Activity	A	B	C	D	E	F	G	H	I	J	K	L
Duration, hours	7	4	6	2	10	6	3	7	5	6	5	3

- (c) Carry out a forward pass and a backward pass through the activity network, showing the early event time and late event time at each vertex of the network.